

(54) Title of the invention : AN IOT-BASED MILK QUALITY ASSESSMENT SYSTEM

		(71)Name of Applicant : 1)PAWAR, Suvarna Eknath Address of Applicant :MIT Art, design and Technology University, Pune, Maharashtra, India ----- 2)PUJERI, Ramachandra 3)MUNDE, Komal 4)KARATH, Shrenuj Santosh 5)CHOPADE, Pradyum Kisan 6)PATIL, Nishant 7)TYAGI, Shanu 8)KULKARNI, Sumant Name of Applicant : NA Address of Applicant : NA (72)Name of Inventor : 1)PAWAR, Suvarna Eknath Address of Applicant :MIT Art, design and Technology University, Pune, Maharashtra, India ----- 2)PUJERI, Ramachandra Address of Applicant :ProVC, MIT Art, design and Technology University, Pune, Maharashtra, India ----- 3)MUNDE, Komal Address of Applicant :MIT Art, design and Technology University, Pune, Maharashtra, India ----- 4)KARATH, Shrenuj Santosh Address of Applicant :MIT Art, design and Technology University, Pune, Maharashtra, India ----- 5)CHOPADE, Pradyum Kisan Address of Applicant :MIT Art, design and Technology University, Pune, Maharashtra, India ----- 6)PATIL, Nishant Address of Applicant :MIT Art, design and Technology University, Pune, Maharashtra, India ----- 7)TYAGI, Shanu Address of Applicant :MIT Art, design and Technology University, Pune, Maharashtra, India ----- 8)KULKARNI, Sumant Address of Applicant :Professor, ReVA University, Bangalore, Karnataka, India --- -----
(51) International classification	:G01N0033040000, G06T0007000000, H04L0067109700, A61B0005110000, A23C0009142000	
(86) International Application No	:NA	
Filing Date	:NA	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number	:NA	
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :
The present invention is related to an IoT-based milk quality assessment system. this portable milk quality assessment system is designed to detect adulteration and analyze key parameters of raw milk. It revolutionizing milk quality assessment by integrating advanced IoT sensors (106) and machine learning techniques (110). the system measures multiple adulteration parameters (101), including fat content, protein content, lactose content, density, pH level, and contaminants such as formalin and urea. key components include specialized modules (102, 103, 104) for parameter analysis, a (E-sense) microcontroller (105) for data processing, and a user-friendly graphical interface (110). Sensory data (107) undergoes pre-processing (108) and is stored in cloud storage (109) for remote access. The system provides real-time results, indicating whether the milk meets quality standards (111) or is adulterated (112). Its portability, comprehensive analysis, real-time feedback, and cost-effectiveness make it ideal for dairy farmers, processors, and regulators, enhancing milk safety and consumer protection.

No. of Pages : 16 No. of Claims : 5